DATA SCIENCE ALLIANCE • TASK 2.3 CHECK-IN

Downtown San Diego Unsheltered Homeless Population: EDA Findings

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SOURCE DATA
Downtown San Diego Partnership (DSDP) Clean & Safe monthly counts

LONGITUDINAL SCOPE
2017–2025 (8 full & partial calendar years, 7 downtown neighborhoods)

ANALYSIS FOCUS
Overall trajectory, seasonality, COVID dip, methodology & spatial dynamics

Overall Longitudinal Trend: Three Distinct Phases (2017-2025)

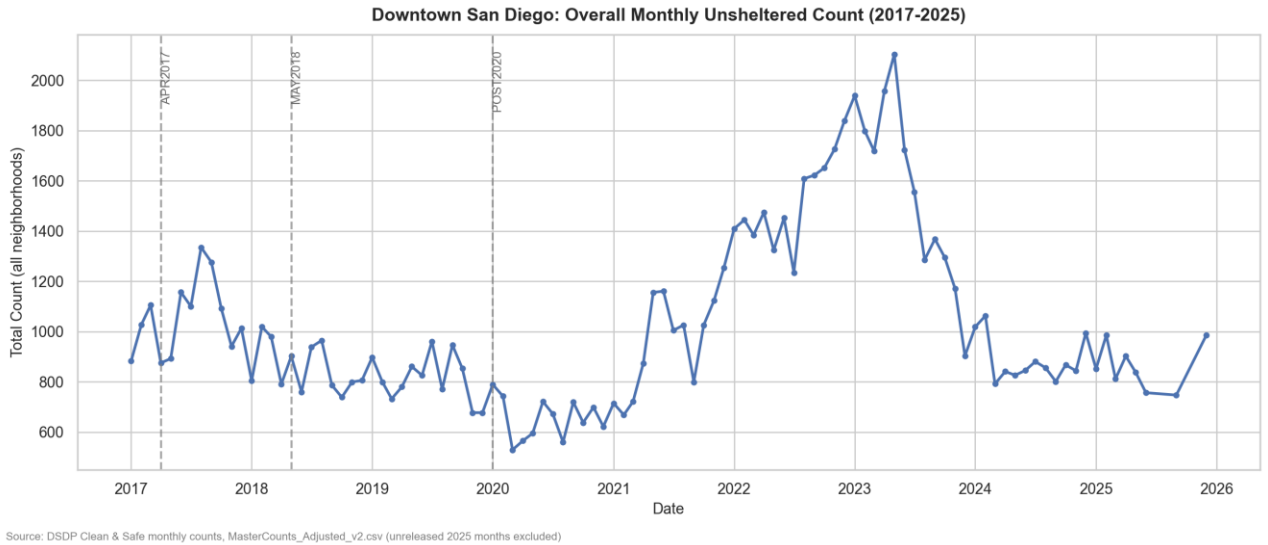
529
SERIES LOW
March 2020 (COVID trough)

2,103
SERIES PEAK
May 2023 (Rebound peak)

156.8
GRAND MEAN
Per neighborhood-month

Trajectory Overview: Three Clear Phases

- **2017–2020 Decline:**
  Annual average fell from 176.31 to 109.02 (~-38% cumulative drop). Year-over-year changes: -18.98% (2018), -4.94% (2019), and -19.71% (2020).
- **2021–2023 Rebound & Peak:**
  Counts rebounded sharply: +25.85% (2021) and +57.63% (2022), reaching an annual average of 223.98 in 2023 — more than double the 2020 trough.
- **2024–2025 Pullback & Stabilization:**
  A steep -43.52% drop in 2024 brought the annual average down to 126.50; 2025 remained roughly flat (-2.94%) at 122.78.

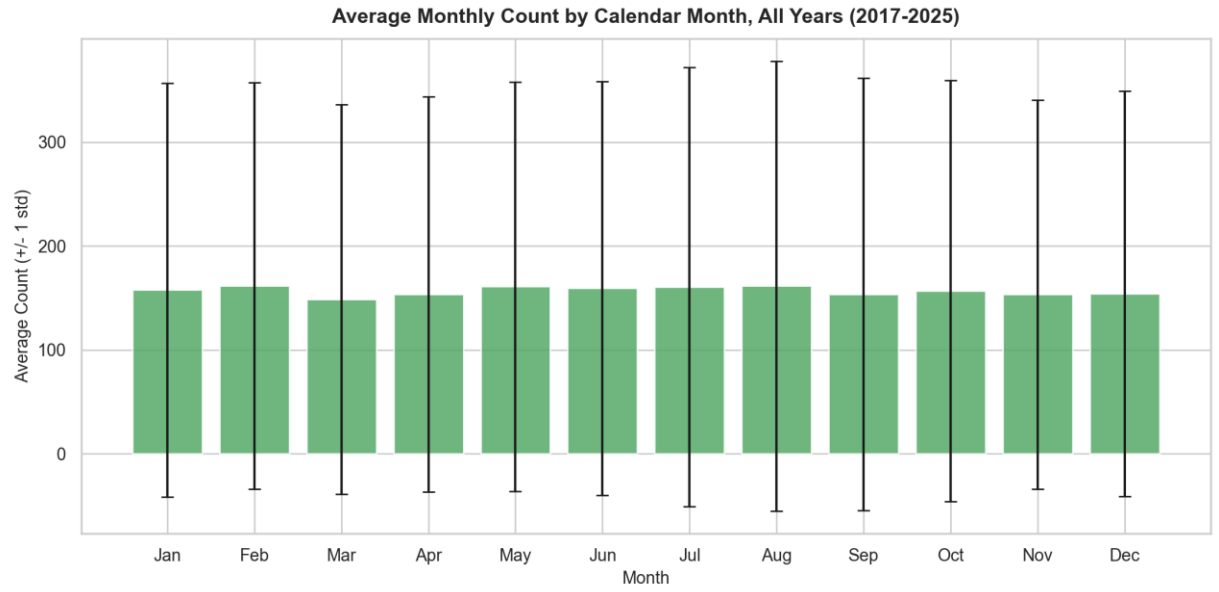


Calendar-Month Seasonality: Modest Variance vs. Annual Swings

Quantifying Calendar Seasonality

Key Takeaway: Which YEAR a month falls in matters far more than which calendar month it is.

- **Modest Range of Deviation:**
  No single calendar month deviates by more than ~5% from the grand mean (156.78). The peak-to-trough spread is only ~8.4 percentage points.
- **Top 3 Highest Months:**
  Feb (+3.2%, 161.76), Aug (+3.1%, 161.61), and May (+2.6%, 160.93).
- **Top 3 Lowest Months:**
  Mar (-5.2%, 148.67), Sep (-2.0%, 153.63), Nov (-2.1%, 153.43) [Apr: -2.1%, 153.53].
- **Large Residual Variance:**
  Month-to-month standard deviations (187–216) dwarf monthly mean differences, confirming macro annual shocks overpower calendar cycles.
- **East Village Sub-Pattern:**
  East Village diverges from citywide rhythm: peaks in Jul–Aug (~601–604) and dips in spring (Mar–Apr, ~509).



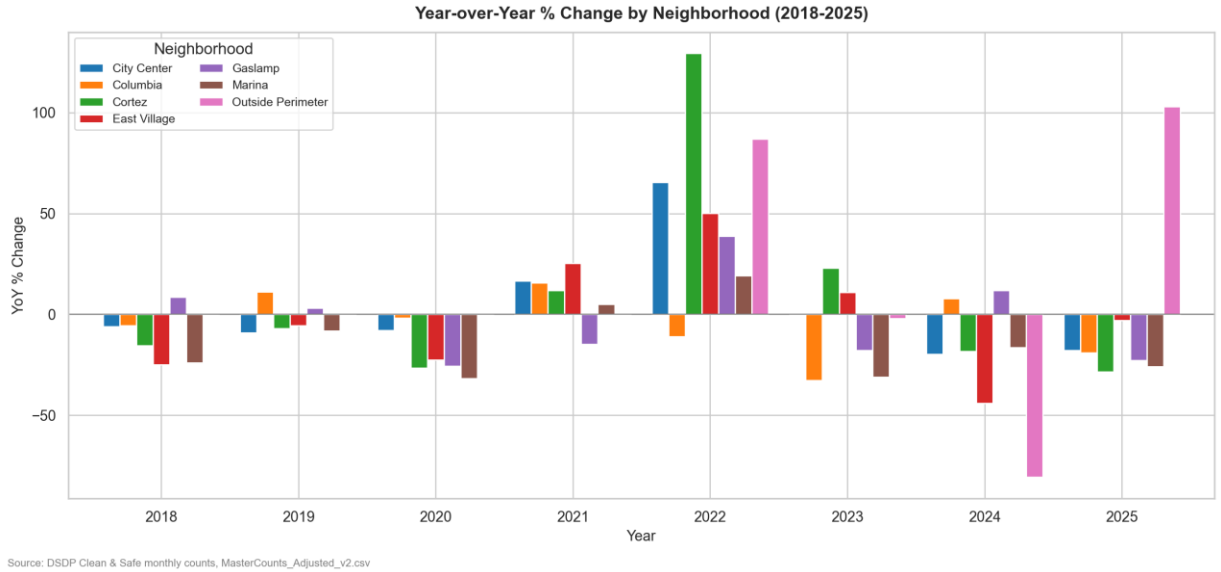
Source: DSDP Clean & Safe monthly counts, MasterCounts_Adjusted_v2.csv

Neighborhood Rankings: Volume Concentration vs. Growth Rates

Volume Scale vs. Growth Dynamic

Core Insight: Growth is NOT concentrated in the largest neighborhood.

- **By Average Monthly Count (Magnitude):**
  1. East Village: 552.77 (std 157.09 — highest volume & volatility)
  2. Outside Perimeter: 217.57 (partial coverage, 56 obs)
  3. City Center: 171.81 (std 54.80)
  4. Cortez: 59.52 | 5. Gaslamp: 48.43
  6. Columbia: 44.49 | 7. Marina: 30.94 (smallest, std 15.82)
- **By Growth Rate (Mean YoY %, 2019+):**
  - Fastest Growing: Outside Perimeter (+26.90%/yr), Cortez (+11.96%/yr), City Center (+3.94%/yr).
  - Modest Core Growth: East Village grew slowly (+1.64%/yr) despite ~55% volume share.
  - Shrinking Areas: Marina (-12.76%/yr) and Columbia (-4.28%/yr) had sustained drops.
- **Spatial Sub-Area Asymmetry:**
  North East & South East EV consistently outcount West sub-areas.

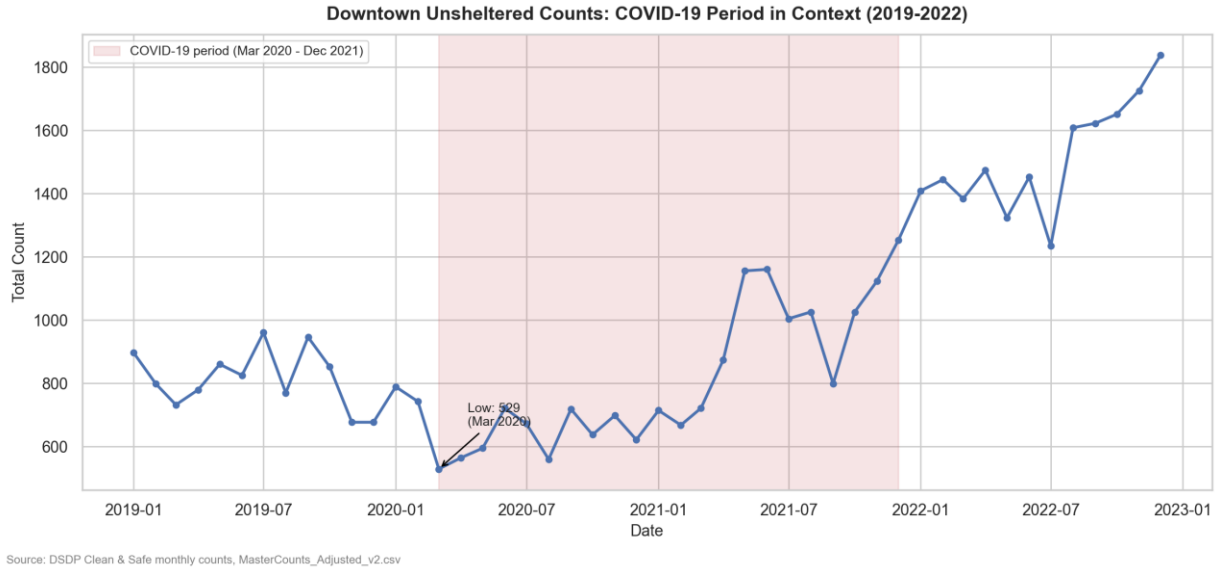


COVID-19 Period Impact: Temporary Suppression & Sharp Rebound

Pandemic Shock & Rebound Profile

Finding: Pandemic programs suppressed visible street counts without a lasting reduction in demand.

- **Series All-Time Low (March 2020):**
  Citywide unsheltered total hit its series low of 529 in March 2020 at the onset of initial COVID shelter-in-place orders.
- **2020 Annual Trough (~20% Reduction):**
  2020 annual average (654.1) dropped ~20% below the 2019 pre-COVID baseline (814.72), reflecting -19.71% YoY contraction.
- **Full Recovery by 2021:**
  Counts recovered to an annual average of 960.37 in 2021 (+25.85% YoY), climbing above pre-COVID baseline levels prior to the 2022–2023 surge.
- **Operational & Policy Mechanisms:**
  Emergency indoor shelter placements (Convention Center Operation Shelter to Home), eviction moratoria, and curtailed street sweeps temporarily reduced visible unsheltered numbers; demobilization preceded the 2022–2023 peak.

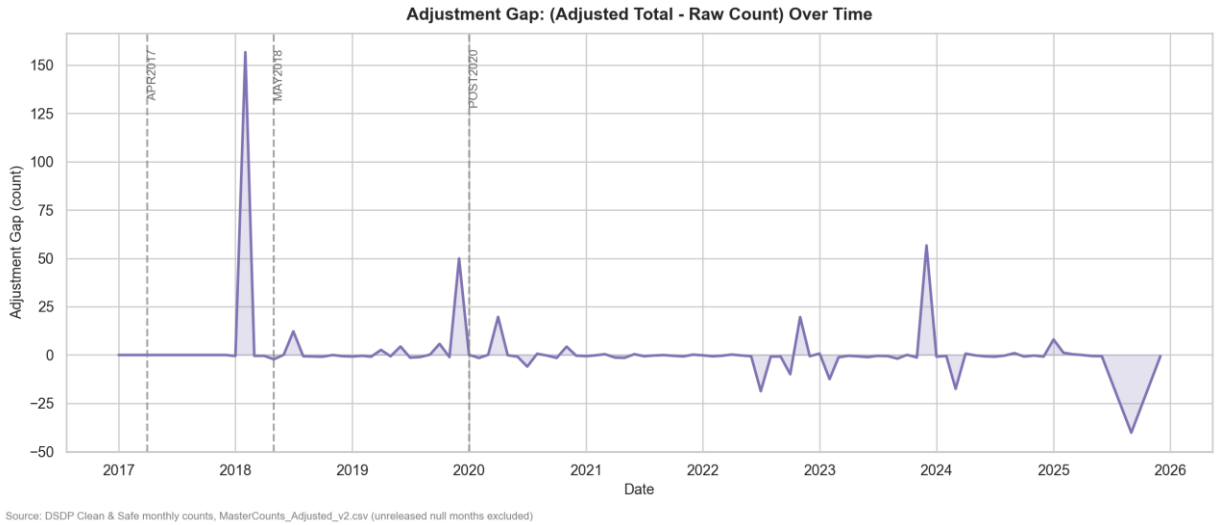


Methodology Regimes: Quantifying the Raw vs. Adjusted Count Gap

Evaluation of Multiplier Regimes

Conclusion: Multipliers produce a proportional gap without distorting longitudinal trendlines.

- **Four Historical Regimes:**
  - PRE2017 (Jan–Mar 2017): Direct visual count, no component breakout.
  - APR2017 (Apr 2017–Apr 2018): Vehicle multiplier set to 1.66×.
  - MAY2018 (May 2018–Dec 2019): Vehicle multiplier increased to 2.03×.
  - POST2020 (Jan 2020–present): 2.03× retained by DSDP for continuity.
- **2018 Volatility & Source Discrepancy:**
  Gap (total_count - raw_count) peaked in 2018 (avg +13.5/mo, spiking to +156.61 in Feb 2018). Audit logged Feb 2018 spike as partly a source-PDF addition error (+30 discrepancy), not purely multiplier mechanics.
- **2019+ Gap Stabilization:**
  From 2019 onward, monthly gap stabilized to near-zero (roughly ±1 to ±5/month), well within expected rounding tolerance (±2).
- **Pre-2018 Constraint:**
  Because component breakouts began in Jan 2018, raw and adjusted totals are identical across all of 2017 (gap = 0).



Hypotheses for Further Analysis: Focus Areas for Later Tasks

H1: Policy & Enforcement Driver of 2022-2023 Peak

SYNCHRONIZED CITYWIDE SURGE

The +57.63% (2022) surge hit nearly every neighborhood simultaneously (Cortez +129%, Outside Perimeter +87%, East Village +50%). This synchronized timing points to systemic citywide drivers (eviction moratorium expirations, shelter demobilizations, or sweep shifts) rather than local dynamics.

- **Test: Correlate monthly counts against RTFH policy timelines, shelter census, and 311 encampment dispatch logs.**

H2: East Village Asymmetry Reflects Built Environment

EAST VS. WEST INFRASTRUCTURE DIVIDE

East Village sub-areas show persistent asymmetry: North East and South East EV consistently outcount North West and South West EV across 2018–2025. This concentration indicates infrastructure and zoning drivers.

- **Test: Spatial autocorrelation (Moran's I / Getis-Ord Gi* in Assignment 5) against transit hubs, shelter proximity, and zoning.**

H3: 2024 Pullback (-43.52%) May Be a Reporting Artifact

RAPID SINGLE-YEAR DROP SCRUTINY

A steep -43.52% drop in a single year immediately after the historical peak warrants verification against external benchmarks before reading it as a true population drop. Count area or staffing changes may be confounders.

- **Test: Benchmark against independent RTFH Point-in-Time (PIT) counts and audit 2024 DSDP canvassing operations.**

H4: Neighborhood Growth Inversely Related to Area Size

PERIPHERAL DISPLACEMENT DYNAMICS

Mid-sized areas grow fastest (Outside Perimeter +26.90%, Cortez +11.96%, City Center +3.94%), while East Village grows slowly (+1.64%) and Marina shrinks (-12.76%). Enforcement in core areas may displace individuals to borders.

- **Test: Perform spatial adjacency and boundary-flow analysis in Assignment 5 to evaluate displacement effects.**

Data Limitations, Governance Caveats & Roadmap Ahead

Identified Data Limitations & Biases

- **Incomplete 2025 Series (Reporting Lag):**
  28 nulls in total_count (Jul, Aug, Oct, Nov 2025 across all 7 neighborhoods) represent unreleased counts, not data errors. All 2025 annual figures must be treated as partial-year.
- **Outside Perimeter Partial History:**
  Outside Perimeter contains only 56 observations (canvassing began ~2022). Its reported +26.90%/yr growth rate is the least statistically reliable neighborhood figure.
- **Pre-2018 Lack of Component Breakouts:**
  Component-level counts (tents, vehicles) are unavailable before Jan 2018. Raw and adjusted totals are identical for all of 2017, precluding early multiplier analysis.
- **Documented Source-PDF Addition Errors:**
  Task 1.1's 45-cell audit found a 97.56% fidelity rate against source PDFs, noting known errors (Feb 2018 +30 PDF addition bug, 2019 EV sub-area duplication, Apr 2025 discrepancy).

Roadmap Ahead: Assignment 3 Integration

- **Immediate Deliverable: Assignment 3:**
  Block-Level Validation & Spatial Integration — moving from macro neighborhood aggregates to granular micro-data across 500+ individual block parcels.
- **Granular Block-Level Mapping:**
  Audit street-level counts, reconcile boundary discrepancies, and align block records with GIS city parcel boundaries.
- **Spatial Autocorrelation Modeling:**
  Execute Global Moran's I and Local Getis-Ord Gi* hot-spot detection in Assignment 5 to quantify spatial clustering around transit nodes and service hubs.
- **Cross-Dataset Validation & Benchmarking:**
  Cross-reference DSDP monthly counts with regional RTFH annual Point-in-Time (PIT) benchmarks and City of San Diego 311 service call logs.